Closing the Deictic Loop

Bidirectional Deictic Grounding for Gesture-Conditioned Vision-Language-Action Navigation on a Quadruped

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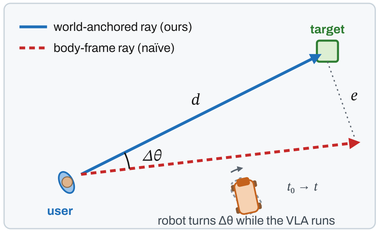
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1. MOTIVATION

Pointing is brief: about 1–2 s, and tied to the pose the robot had when it saw it.

VLA inference is slow. If the robot is still turning and keeps the ray in its body frame, the ray ends up wrong by exactly the rotation $\Delta\theta$ made during inference — a geometric identity, not noise.

Research gap. No existing system combines pointing input, VLA-driven quadruped locomotion and a robot→human grounding channel.

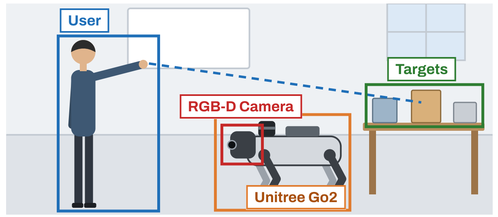


Un-anchored error ≡ $|\Delta\theta|$; miss $e \approx d \cdot \tan\Delta\theta$. LEGS-POMDP: 14.4° ≈ 1.3 m off at 5 m.

2. CONTRIBUTIONS

1. **Geometric identity:** an un-anchored pointing ray is off by exactly $|\Delta\theta|$, the rotation during inference.
2. **Gesture memory buffer:** rays anchored in the world frame (TF2) at capture, re-projected at use.
3. **Bidirectional grounding:** the robot turns, asks, and takes a confirm or a correction.
4. **Gesture-grounding metrics** with a 1–5 s latency sweep.

3. DATA COLLECTION SETUPS



(a) Lab setup — Unitree Go2 with RGB-D camera, Velodyne LiDAR and Jetson AGX Orin; user points from distance d



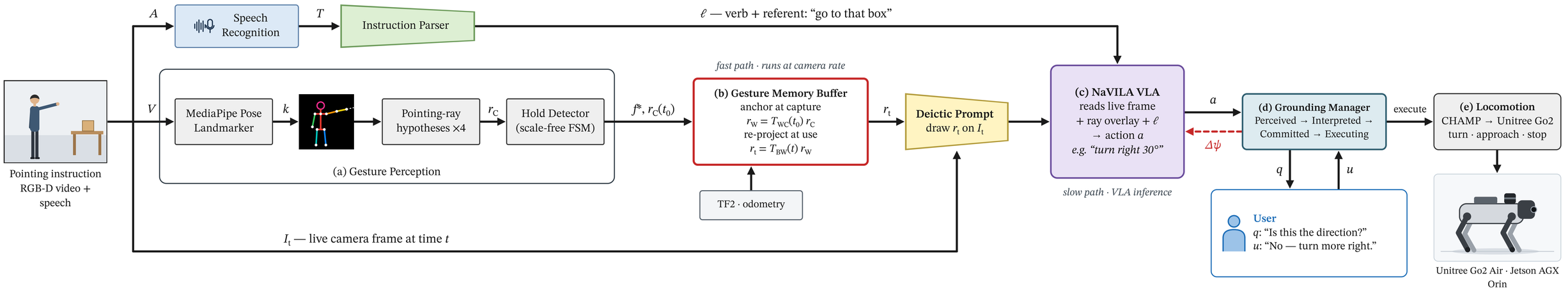
(b) Simulation — Gazebo + CHAMP + HuNavSim pedestrians

(c) Logged per trial

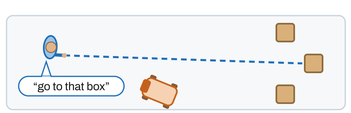
- RGB-D frames V
- Speech audio A
- Pose keypoints k
- TF2 / odometry
- Velodyne LiDAR
- GT target pose

Sweeps: distance · target layout · injected latency τ = 1–5 s

4. PROPOSED METHOD — SYSTEM ARCHITECTURE

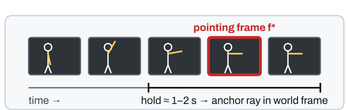


5. FROM POINTING TO APPROACH



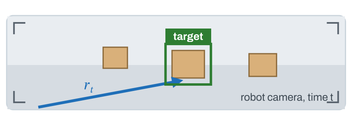
1 Point & speak

User points at a box and says "go to that box".



2 Perceived

Hold detected; pointing frame f^* kept, ray anchored in the world frame.



3 Interpreted

NaVILA reads the live frame with r_t drawn in and picks the box.



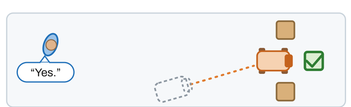
4 Committed

Robot turns to face its pick and asks "Is this the direction?"



5 Corrected

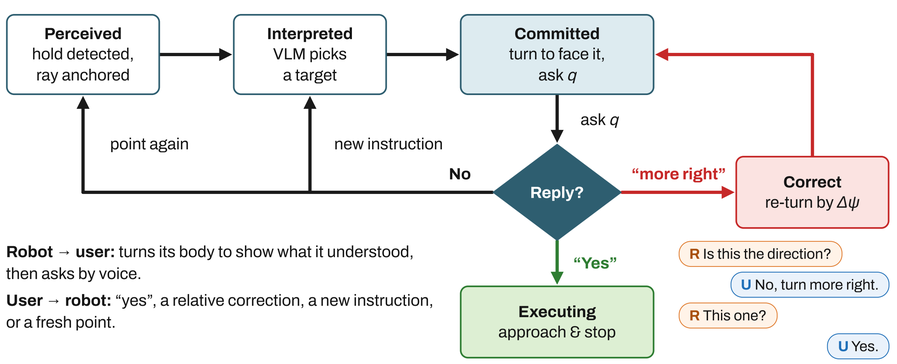
"No, turn more right." Robot adds $\Delta\psi$, re-turns and asks again.



6 Executing

"Yes." Robot walks to the box and stops in front of it.

6. BIDIRECTIONAL GROUNDING LOOP



Robot → user: turns its body to show what it understood, then asks by voice.

User → robot: "yes", a relative correction, a new instruction, or a fresh point.

7. EVALUATION

Deictic heading error
final heading vs true target bearing (°)

Instruction capture rate
pointing events caught (%)

Target selection accuracy
commits to the right object (%)

Latency-conditioned degradation
all metrics vs injected τ = 1–5 s

Compare: body-frame ray (naïve) vs world-anchored ray (ours).

Where: HuNavSim in Gazebo, then the real Go2.

Platform Unitree Go2 Air · Jetson AGX Orin 32 GB · Velodyne LiDAR · RGB-D camera · ROS 2 Humble · TF2 · MediaPipe · NaVILA · CHAMP · Gazebo · HuNavSim

Related work NaVILA · GIVE · LEGS-POMDP · GesVLA · TIC-VLA · HuNavSim